How the Antennas Are Arranged in Each Configuration

ULA, USA, ECA — and why "local" and "CAPE" are *not* new arrangements

Explanatory note (simple English)

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The most important thing first. In your study there are really only *three* physical antenna arrangements: the **dense ULA**, the **uniform sparse (USA)**, and the **ECA** (and a smaller ECA used as a control). The names you see in the figures — "ULA + local", "ULA + CAPE", "ECA + local", "ECA + CAPE" — are **not four different arrangements**. They are two arrangements (ULA and ECA), each used with two different *processing methods* ("local" or "CAPE"). *The antennas do not move when you switch from local to CAPE.*

1 Two separate things: arrangement vs. processing

Antenna arrangement (hardware: where the antennas sit)	**Processing method** (software: what we do with the received signals)
ULA — dense, equal spacing	**local** — each AP estimates its own channel by itself
USA — sparse, equal spacing	**CAPE** — the APs cooperate to find the user's position first
ECA — sparse, *unequal* spacing	

A "scheme" in your figures is one arrangement *plus* one processing method:

Scheme in the figures	Antennas (hardware)	Processing (software)	Role
ULA + local	dense ULA (64)	local	your original baseline
ECA + local	ECA (63)	local	plain sparse array (Code C)
ULA + CAPE	dense ULA (64)	CAPE	control / ablation
ECA + CAPE (proposed)	ECA (63)	CAPE	**the proposed system**

Read it as "**arrangement + processing**". "ULA + local" and "ULA + CAPE" are the *same antennas*, processed two different ways. "ECA + local" and "ECA + CAPE" are the *same antennas*, processed two different ways.

2 The antenna arrangements (hardware)

All arrays use about the same number of antennas (≈ 64). The spacing unit below is $d_0 = \lambda/2 = 5\,\text{cm}$ (half a wavelength at 3 GHz). Each arrangement just decides *where* the antennas sit on the line.

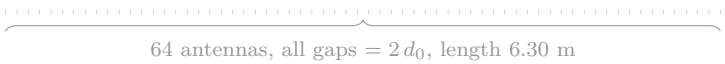
2.1 Dense ULA — equal small spacing

Every antenna is exactly one step (d_0) from the next. 64 antennas, total length 3.15 m. Short array ⇒ poor ability to focus in distance, but no ghost beams.



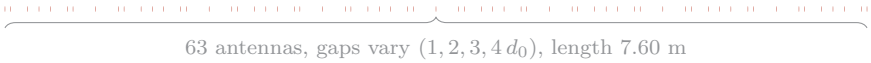
2.2 Uniform sparse (USA) — equal *large* spacing

Same 64 antennas, but every gap is two steps ($2d_0 = \lambda$). The array is now twice as long (6.30 m), so it is sharper — *but* equal large spacing creates exact copies of the beam at other angles (grating lobes / ghost beams). This is why USA is not used in the end.



2.3 ECA — sparse with *unequal* spacing (the proposed array)

63 antennas placed at *uneven* gaps, so the array is the widest (7.60 m) and sharpest, yet the uneven gaps stop the ghost copies from lining up — no strong ghost beams.



How the ECA gaps are built (the co-prime recipe). Take two numbers with no common factor: $M = 5$ and $N = 4$. One "block" is made of two rulers that share their first antenna:

- a 5-antenna ruler with gap $4\, d_0$: positions $0, 4, 8, 12, 16$;
- a 4-antenna ruler with gap $5\, d_0$: positions $0, 5, 10, 15$.

Merging them (position 0 is shared) gives 8 antennas at $\{0, 4, 5, 8, 10, 12, 15, 16\}$. This block is repeated 8 times (period $20\, d_0$) and mirrored about the centre, giving 63 antennas. The gap pattern that repeats along the array is

$$\ldots,\ 1,\ 3,\ 2,\ 2,\ 3,\ 1,\ 4,\ 4,\ \ldots \quad \text{(in units of } d_0\text{)}.$$



2.4 ECA same-aperture (the small control)

The same ECA recipe but with only 4 blocks instead of 8: 31 antennas over 3.60 m (about the dense ULA's length). It is in the study only to separate "more aperture" from "sparser spacing"; it is not the proposed array.

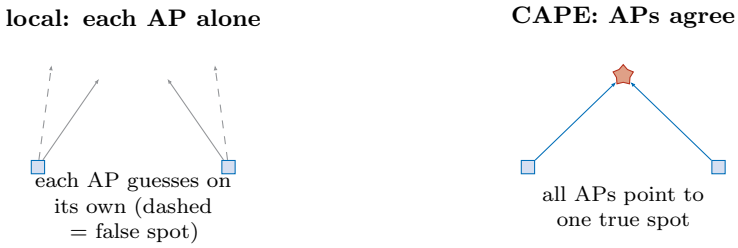
Arrangement	Antennas	Gaps	Length	Ghost beams
Dense ULA	64	all $1\, d_0$	3.15 m	none
Uniform sparse USA	64	all $2\, d_0$	6.30 m	**strong**
ECA (proposed)	63	$1, 2, 3, 4\, d_0$	**7.60 m**	none (suppressed)
ECA same-aperture	31	$1, 2, 3, 4\, d_0$	3.60 m	none (control)

3 The processing methods (software) — same antennas

These do *not* change the antenna positions. They change only how the central processor turns the received pilot signals into a channel estimate.

"local" estimation. Each AP looks at its own antennas only and guesses the user's channel by itself. Fast and simple. Problem on a sparse array: the AP sees several "maybe" spots (ambiguity) and sometimes picks the wrong one, so the estimate is noisier. This is the old method (your baseline and Code C).

"CAPE" estimation (proposed). The APs *cooperate*. A user has only one true location; the strongest APs propose candidate positions, the central processor keeps the one position that all of them agree on (the false spots of one AP are killed by the others), and then every AP builds its channel from that single agreed position. Same antennas, smarter use of them.



Why ULA + CAPE is in the study but is a bad idea. CAPE needs a sharp position fix, which only a *wide* array gives. The dense ULA is short, so its position fix is poor; forcing CAPE on it makes things worse (its curves floor out). So CAPE belongs with the *sparse* array. "ULA + CAPE" is kept only to prove this point — that the array and the processing must go together. The winning pair is **ECA + CAPE**.

4 One-page summary

- **ULA**: 64 antennas, equal $\lambda/2$ gaps, short (3.15 m). Simple, low resolution.
- **USA**: 64 antennas, equal λ gaps, long (6.30 m). Sharp but has ghost beams.
- **ECA**: 63 antennas, *unequal* gaps (1–4 d_0), longest (7.60 m). Sharp, no ghosts — the proposed array (from the 2025 ECA paper).
- **"local" and "CAPE" are processing, not arrangements.** The antennas are identical; only the way the signals are combined changes.
- **ECA + CAPE** = the proposed system: the widest, sharpest array, made reliable by letting the APs agree on the user's position.